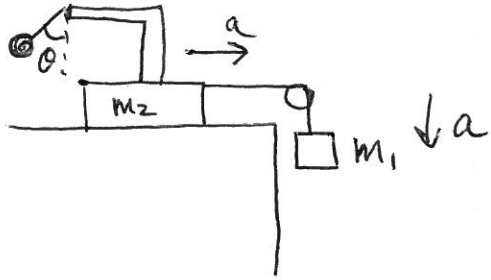
MidTerm 2 Solutions 10/23/09

P1



$m_1 = 250$ kg
$m_2 = 1250$ kg

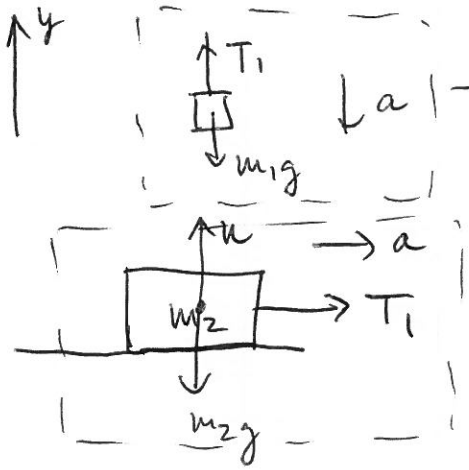
a) let us start by finding relation between angle θ and acceleration a:

$$\begin{cases} T\cos\theta - mg = 0 \Rightarrow T = \dfrac{mg}{\cos\theta} \\ T\sin\theta = ma \end{cases}$$

$$\Rightarrow \frac{mg}{\cos\theta}\sin\theta = ma$$

$$a = g\frac{\sin\theta}{\cos\theta} = g\tan(\theta).$$

Now find a if no friction is present:



$$T_1 - m_1 g = -m_1 a$$
$$T_1 = m_1(g-a)$$

$$T_1 = m_2 a$$

$$\Rightarrow m_1(g-a) = m_2 a \Rightarrow (m_1+m_2)a = m_1 g$$

$$a = \frac{m_1}{m_1+m_2} g$$

$\Rightarrow$ answer for part (b)

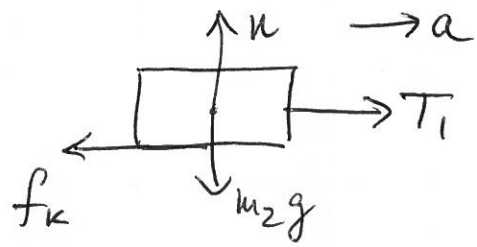
$$\tan\theta = \frac{a}{g} = \frac{m_1}{m_1+m_2} = \frac{250}{1500} = \frac{1}{6}$$

$$\Rightarrow \theta = 9.46^\circ$$

c) $\tan\theta = \dfrac{m_1}{m_1+m_2} = \dfrac{1}{1+\frac{m_2}{m_1}}$

The largest angle is achieved when $\frac{m_2}{m_1} \ll 1$ so that $\tan(\theta) = 1 \Rightarrow \theta = 45°$.

d) Friction is present: re-do free body diagram for $\underline{m_2}$



$$\begin{cases} T_1 - f_k = m_2 a \\ n - m_2 g = 0 \;, \; f_k = \mu_k n = \mu_k m_2 g \end{cases}$$

$$\Rightarrow T_1 = m_1(g-a) = \mu_k m_2 g + m_2 a$$

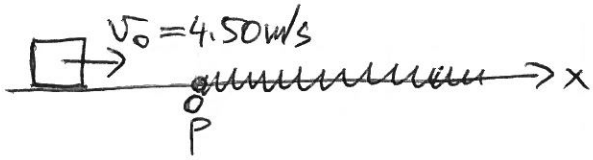
$$(m_1 + m_2)a = m_1 g - \mu_k m_2 g$$

$$a = \frac{m_1 - \mu_k m_2}{m_1 + m_2} g$$

$$\Rightarrow \tan\theta = \frac{m_1 - \mu_k m_2}{m_1 + m_2} = \frac{250 - 0.2 \cdot 1250}{1500} = 0$$

$\Rightarrow \theta = 0$ as acceleration $a = 0$.

P2



a) $K_1 - K_0 = W_{TOT}$; $\mu_k = 0.100$

$$W_{TOT} = W_{friction} + \underset{\substack{\parallel \\ 0}}{W_n} + \underset{\substack{\parallel \\ 0}}{W_g}$$

displacement $\perp$ force

$$W_{friction} = -\mu_k n x \quad , n = mg$$

Box stops when $v_1 = 0 \Rightarrow K_1 = \frac{1}{2} m v_1^2 = 0$

$$\Rightarrow -\frac{1}{2} m v_0^2 = -\mu_k n x$$

$$x = \frac{v_0^2}{2\mu_k g} = 10.3 \text{ m}$$

b) Now $\mu_k(x) = 0.1 + \alpha x$

$$\Rightarrow W_{friction} = -\int_0^x \mu_k(x) n \, dx$$

$$\left(\text{recall } W_F = \int \vec{F} \cdot d\vec{\ell} \right)$$

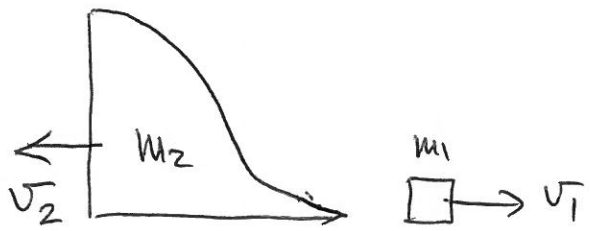
$$W_{friction} = -\int_0^x (0.1 + \alpha x) m_2 g \, dx =$$

$$= -m_2 g \left(0.1 x + \frac{\alpha}{2} x^2\right)$$

$$\Rightarrow -\frac{1}{2} m v_0^2 = -m_2 g \left(0.1 x + \frac{\alpha}{2} x^2\right)$$

$$\frac{v_0^2}{2g} = 0.1 x + \frac{0.04}{2} x^2 \Rightarrow x = 5.11 \text{ m}$$

P3



$v_1 = 4.00\ m/s$

a) Initial momentum $p = 0$

Final $p = -m_2 v_2 + m_1 v_1$

$$\Rightarrow \quad 0 = -m_2 v_2 + m_1 v_1$$

$$v_2 = \frac{m_1}{m_2} v_1 = \frac{0.500}{3.00} 4.00\ m/s = \frac{2}{3}\ m/s$$

$$= 0.667\ m/s$$

b) If no friction present $E_i = m_1 g h$

$$E_f = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

$$\Rightarrow \quad m_1 g h = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

$$h = \frac{v_1^2}{2g} + \frac{m_2}{m_1} \frac{v_2^2}{2g} = \frac{v_1^2}{2g}\left(1 + \frac{m_1}{m_2}\right) =$$

$$= 0.952\ m$$

P4

$m_1 = 0.008$ kg
$m_2 = 0.492$ kg
$k = 5.00$ N/m
$v_1 = 325$ m/s

Conservation of momentum:

before collision $p = m_1 v_1$

right after $p = (m_1 + m_2) v_2$

$\Rightarrow \quad (m_1 + m_2) v_2 = m_1 v_1$

$$v_2 = \frac{m_1 v_1}{m_1 + m_2} = \frac{0.008}{1.00} 325 \text{ m/s} = 2.60 \text{ m/s}$$

Max compression: Kinetic energy of block + bullet

$$K = \frac{1}{2}(m_1 + m_2) v_2^2$$

Elastic energy of spring

$$U = \frac{1}{2} k x^2$$

$$\frac{1}{2} k x^2 = \frac{1}{2}(m_1 + m_2) v_2^2$$

$$x = \sqrt{\frac{(m_1 + m_2) v_2^2}{k}} = 1.16 \text{ m}$$